CS-104 Retrieval Quiz (ANSWER KEY)

Fall 2026 · Week 4, Wednesday

Name: _____

Closed book, no devices. For questions 1–3, circle the one best answer. For question 4, write your code in the space provided (no algorithm, documentation, or test cases needed).

1.

What does this code print?

```
x = 95
if x >= 90:
    print("You've got an A")
if x >= 80:
    print("You've got a B")
else:
    print("Keep studying!")
```

a) Only `You've got an A`

b) Only `You've got a B`

c) `You've got an A` **and then** `You've got a B` **[correct]**

d) `You've got an A` and then `Keep studying!`

2.

Which expression is **not** equivalent to `13 <= age < 20`?

a) `13 <= age and age < 20`

b) `age >= 13 and age < 20`

c) `not (age < 13 or age >= 20)`

d) `13 < age <= 20` **[correct]**

3.

What does this code print?

```
a = 5
result = (a == 3 or 4)
if result:
    print("yes")
else:
    print("no")
```

a) `yes`**, because** `False or 4` **gives** `4`**, which is truthy [correct]**

b) `no`, because a is not 3

c) Nothing; it raises a `TypeError`

d) Nothing; neither branch runs

4.

Write a code segment that reads a pH value (a float) from the user and prints `error` if it is outside the range 0 to 14. Otherwise it prints `acid` if the pH is below 7, `base` if it is above 7, and `neutral` if it is exactly 7.

Sample solution:

```
ph = float(input("pH: "))
if ph < 0 or ph > 14:
    print("error")
elif ph < 7:
    print("acid")
elif ph > 7:
    print("base")
else:
    print("neutral")
```